EXPLORATORY ANALYSIS AND CUSTOMER SEGMENTATION USING UNSUPERVISED MACHINE LEARNING

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INTRODUCTION

Customer segmentation is an important application of unsupervised machine learning that helps businesses understand customer behaviour and identify groups with similar characteristics. By grouping customers into meaningful segments, organisations can improve marketing strategies, customer targeting, and personalised services.

This project analyses a customer shopping behaviour dataset containing demographic and spending-related information, including gender, age, annual income, and spending score. Exploratory data analysis, dimensionality reduction, and clustering techniques were applied to identify hidden customer patterns and evaluate clustering performance.

Two clustering algorithms, K-Means and DBSCAN, were implemented and compared using multiple internal evaluation metrics, including Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. The overall objective of the study was to determine the most effective clustering approach for customer segmentation.

The analysis was conducted in Python using Jupyter Notebook. Pandas and NumPy were used for data preprocessing and descriptive statistics, while Matplotlib and Seaborn were used for visualisation. Scikit-learn was used for dimensionality reduction and clustering implementation, including PCA, t-SNE, UMAP, K-Means, and DBSCAN.

The identified customer clusters also provide useful business insights. High-income high-spending customers may represent premium customer segments with strong commercial value, while low-income high-spending customers may be more responsive to promotions and trend-based marketing strategies. Similarly, high-income low-spending customers may represent potential upselling opportunities. These findings demonstrate the practical business value of customer segmentation using clustering techniques.

2. EXPLORATORY DATA ANALYSIS

The dataset contains 200 customer records with demographic and behavioural attributes including gender, age, annual income, and spending score. Initial exploratory analysis was performed to understand feature distributions, identify possible outliers, and analyse relationships between variables.

The gender distribution was relatively balanced between male and female customers. Histograms and kernel density estimation plots showed that age and annual income followed moderately spread distributions, while spending scores demonstrated larger variation across customers. This indicates the existence of customers with significantly different purchasing behaviours.

Boxplots revealed several potential outliers, particularly within annual income and spending score features. However, these observations were retained because they may represent meaningful customer groups rather than data errors.

Correlation analysis using Pearson and Spearman heatmaps showed weak linear relationships between most variables. Annual income and spending score exhibited low correlation, suggesting that high-income customers do not necessarily spend more. This justified the use of clustering techniques to uncover hidden behavioural patterns not observable through simple correlation analysis.

The pairplot visualisation further confirmed the presence of separable customer groups, particularly when comparing annual income and spending score. These observations suggested that clustering algorithms may successfully identify distinct customer segments.

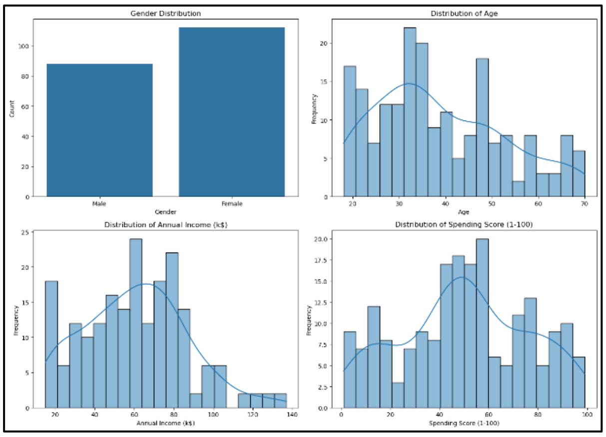

Figure1 Data Distribution Visualization

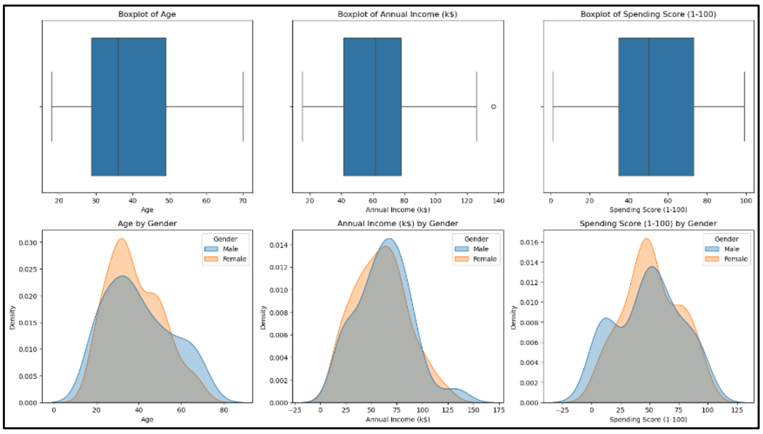

Figure 2 Boxplots and Gender-Based Distribution Analysis

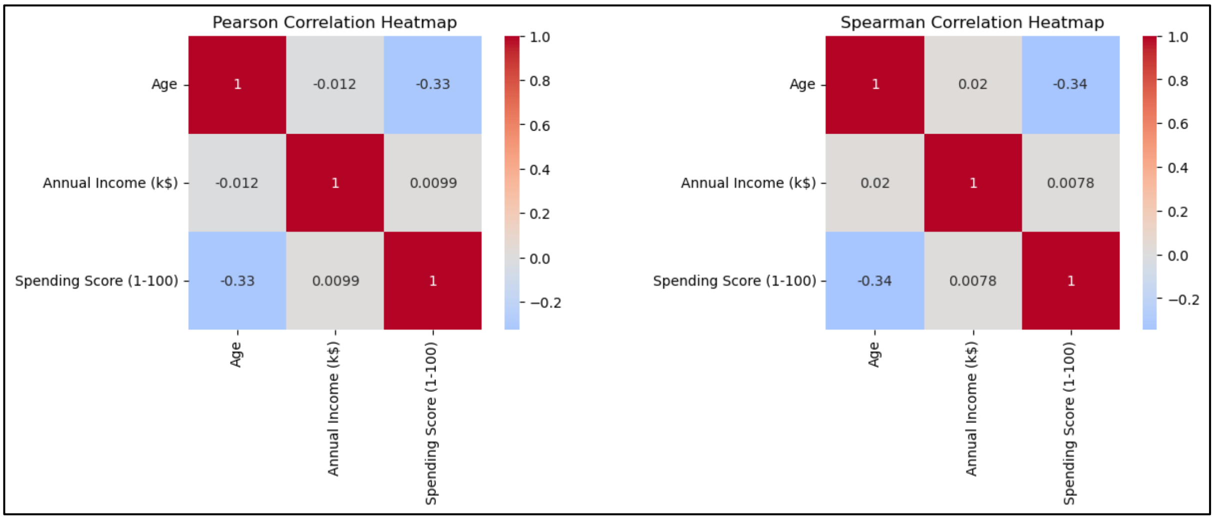

Figure 3 Correlation Analysis

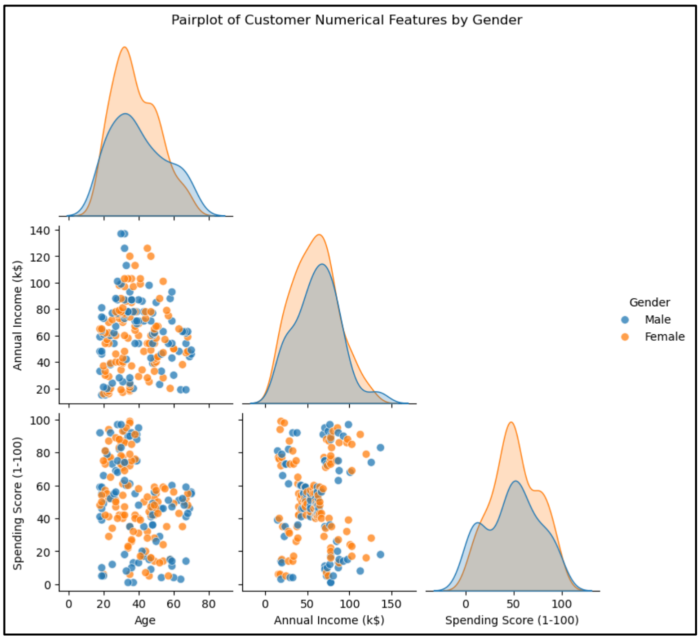

Figure4 Pairwise Relationship Analysis

3. DIMENSIONALITY REDUCTION ANALYSIS

Principal Component Analysis (PCA) was applied to reduce dimensionality and analyse feature variance. The scree plot showed that the first two principal components captured most of the dataset variance, indicating that lower-dimensional representations were sufficient for visual exploration.

To better understand hidden data structures, non-linear dimensionality reduction techniques including t-SNE and UMAP were also applied. Both visualisations revealed visible customer grouping patterns, especially when coloured according to spending score and gender.

The UMAP visualisation showed clearer separation between customer groups compared to PCA and t-SNE, suggesting that the dataset contains non-linear relationships suitable for clustering analysis. These findings supported the application of K-Means and DBSCAN clustering algorithms in the next stage of the project.

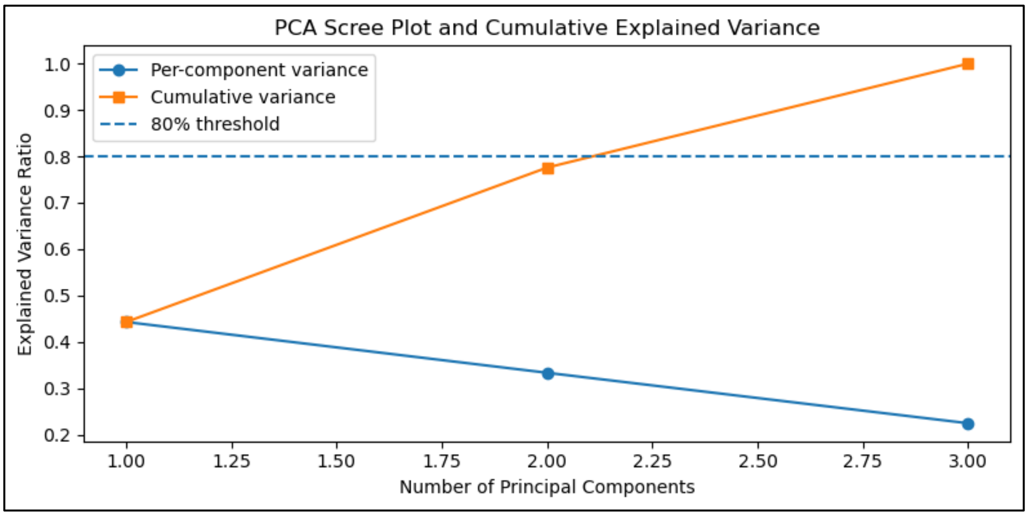

Figure 5 PCA scree plot

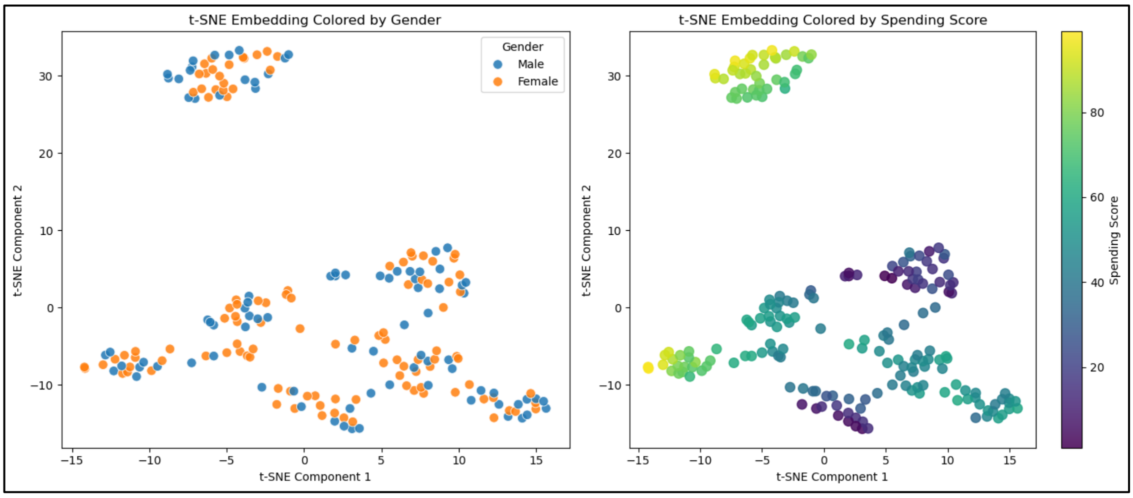

Figure 6 t-SNE visualization

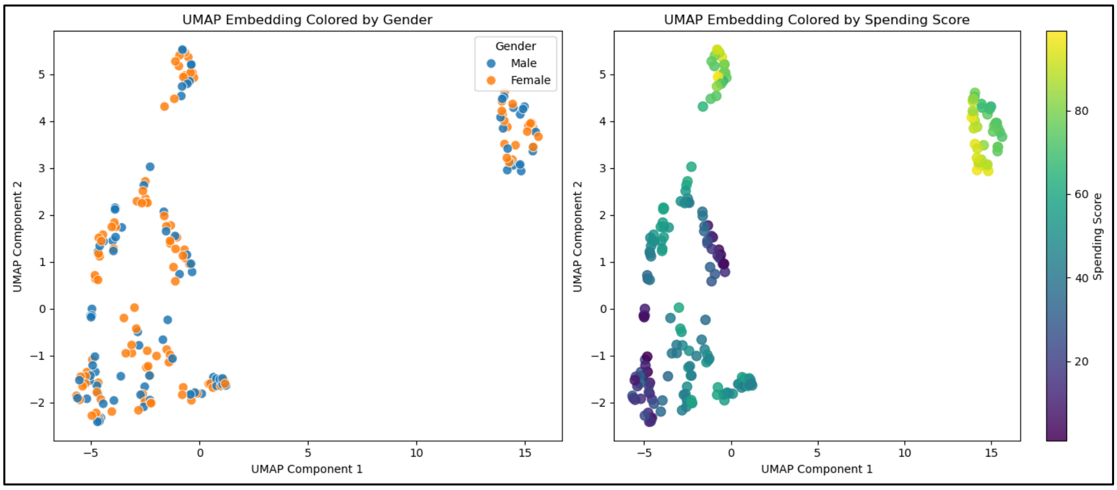

Figure 7 UMAP visualization

4. CLUSTERING ALGORITHM DESIGN

Customer segmentation is an unsupervised learning problem because the dataset does not contain predefined labels indicating customer groups. Clustering algorithms are therefore suitable for identifying hidden patterns and grouping customers based on similar behavioural characteristics.

K-Means clustering was selected because it is efficient, widely used, and performs well on compact and relatively spherical clusters. Since the exploratory analysis and dimensionality reduction visualisations suggested the existence of separable customer groups, K-Means was considered an appropriate initial clustering approach.

DBSCAN was additionally selected because it is density-based and capable of identifying irregular cluster shapes and noise points. Unlike K-Means, DBSCAN does not require the number of clusters to be specified beforehand. This provides an alternative perspective on the customer structure and allows comparison between centroid-based and density-based clustering methods.

The exploratory analysis from Task 1 strongly influenced clustering design decisions. Correlation analysis indicated weak relationships between variables, while t-SNE and UMAP visualisations revealed possible hidden cluster structures. Annual income and spending score were selected as the primary clustering features because they showed the clearest separation patterns during exploratory analysis.

5. CLUSTERING RESULTS AND EVALUATION

After applying K-Means and DBSCAN clustering algorithms, multiple evaluation metrics and visualisation techniques were used to assess clustering quality. The results showed that both algorithms successfully identified meaningful customer groups based on shopping behaviour. K-Means produced clear and well-separated customer segments, while DBSCAN additionally identified noise points and dense customer regions. Overall, DBSCAN achieved slightly better clustering performance based on Silhouette Score and Davies-Bouldin Index, although both algorithms performed effectively on the dataset. Visualisations including cluster heatmaps and 3D segmentation plots further confirmed the presence of distinct customer behaviour patterns.

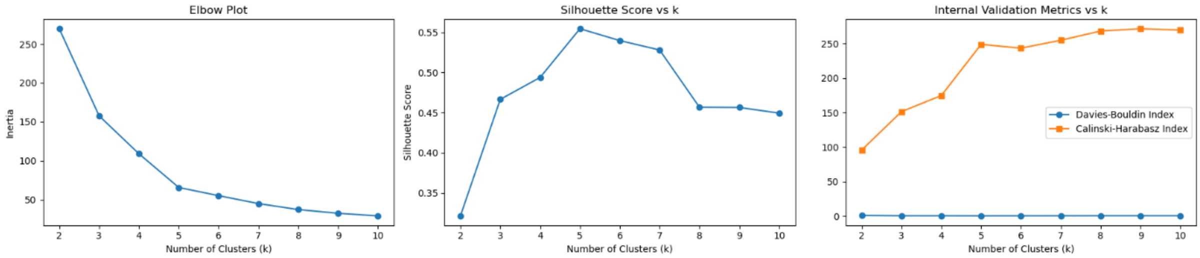

Figure 8: KMeans Evaluation Metrics

Figure 8 shows the elbow method results for K-Means clustering. The inertia value decreased rapidly until approximately k=5, after which the rate of improvement became smaller. This suggests that five clusters provide a suitable balance between cluster compactness and model complexity.

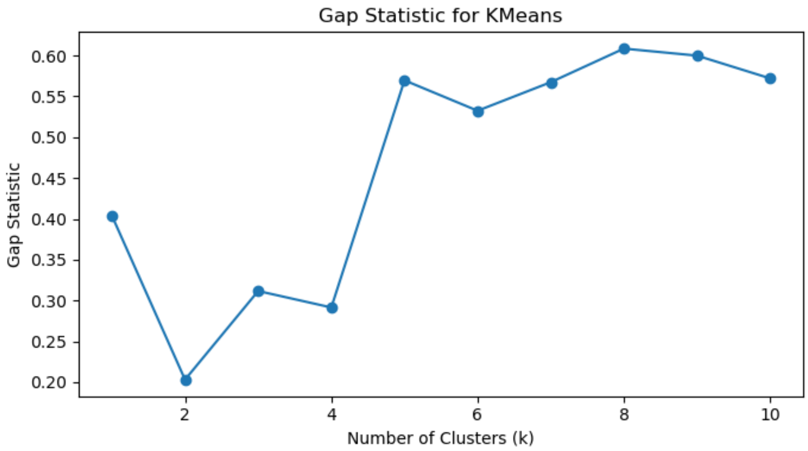

Figure9 Gap Statistic

The gap statistic analysis shown in Figure 9 further supported the selection of five clusters. Higher gap values indicate stronger clustering structure compared with randomly distributed reference data. The results confirmed that the customer data contains meaningful grouping patterns suitable for clustering analysis.

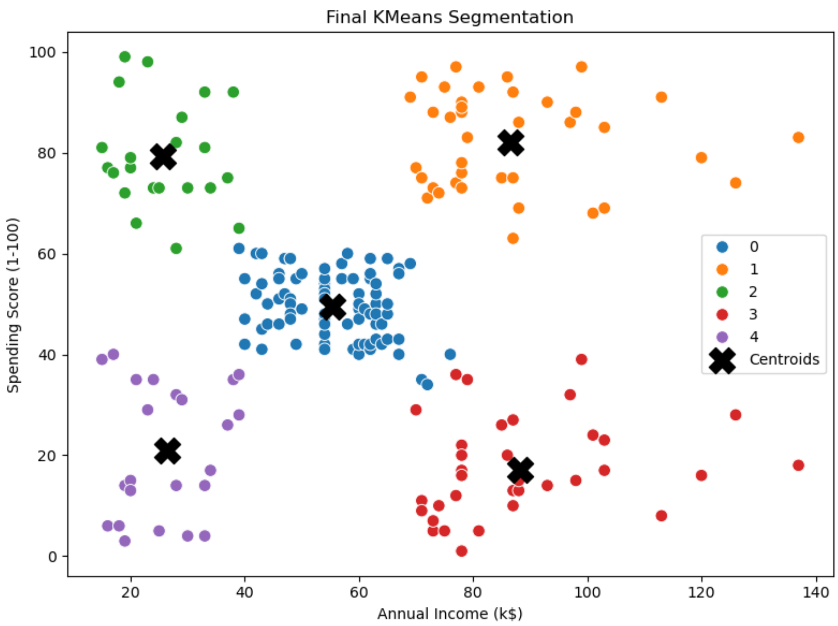

Figure 10 Final KMeans Segmentation

Figure 10 illustrates the final K-Means customer segmentation using annual income and spending score. The algorithm successfully separated customers into distinct behavioural groups, including high-income high-spending customers, low-income low-spending customers, and moderate customer segments. The cluster centroids clearly represent the centre of each customer group.

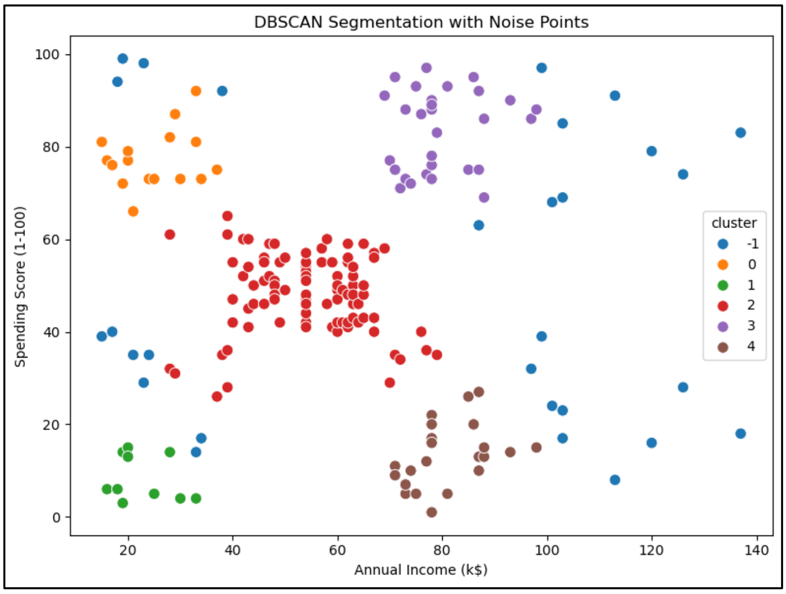

Figure 11 DBSCAN Segmentation

Figure 11 presents the DBSCAN clustering results. Unlike K-Means, DBSCAN identified dense customer regions while additionally detecting several noise points. This demonstrates the ability of DBSCAN to identify irregular cluster structures and outliers without requiring a predefined number of clusters.

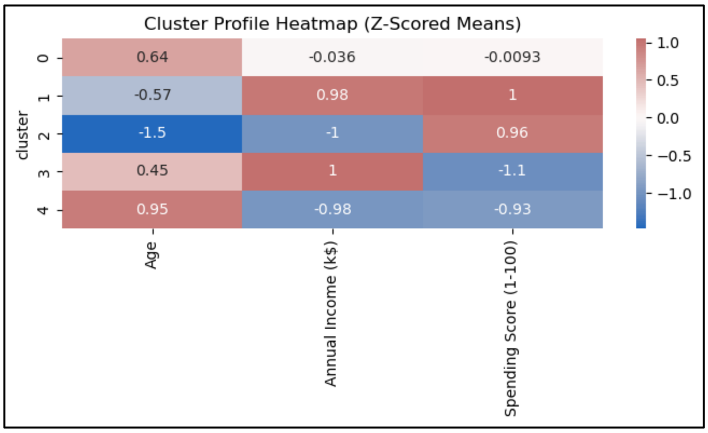

Figure 12 cluster heat map

The cluster profile heatmap in Figure 12 highlights behavioural differences between customer groups. Certain clusters demonstrated higher spending scores and annual incomes, while others represented lower-spending customer groups. The heatmap confirms that the clustering algorithms identified meaningful and interpretable customer segments.

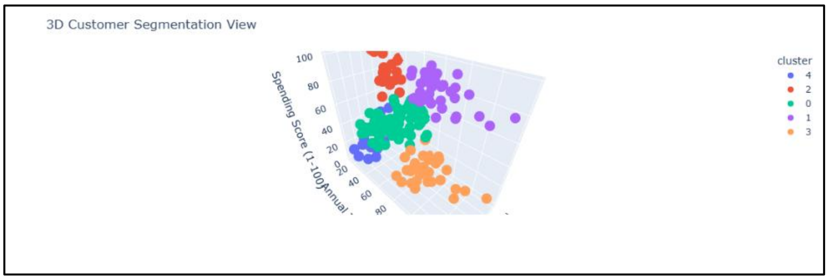

Figure 13 3D Visualization

The 3D customer segmentation visualisation shown in Figure 13 provides additional insight into the separation between customer groups across age, annual income, and spending score simultaneously. The visualisation demonstrates that the identified clusters remain relatively distinct when analysed across multiple dimensions.

	Model	Clusters	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Index
0	KMeans	5	0.555	0.572	248.649
1	DBSCAN	5	0.574	0.483	245.144

Figure 1 model comparison

Table compares the performance of K-Means and DBSCAN using multiple internal evaluation metrics. DBSCAN achieved a slightly higher Silhouette Score (0.574) and lower Davies-Bouldin Index (0.483), indicating better cluster compactness and separation. K-Means achieved a marginally higher Calinski-Harabasz Index, suggesting strong overall cluster structure. Since the performance differences were relatively small, both algorithms performed effectively on the dataset. However, DBSCAN demonstrated slightly better overall clustering quality.

6. CONCLUSION

This project successfully applied exploratory data analysis, dimensionality reduction, and clustering techniques to perform customer segmentation on shopping behaviour data. Visual analysis and dimensionality reduction methods revealed meaningful hidden structures within the dataset, supporting the application of clustering algorithms.

Both K-Means and DBSCAN produced effective customer segments with strong evaluation scores. DBSCAN demonstrated slightly better overall clustering performance, although both algorithms generated meaningful and interpretable customer groups.

The findings demonstrate how unsupervised machine learning techniques can support customer understanding and behavioural analysis for business decision-making.

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